

Vivekanand Dhavali

Data Scientist · Hospitality & E-commerce Analytics

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PROFESSIONAL SUMMARY

Results-oriented Data Scientist with 5+ years of cross-functional experience in data analysis, integration systems, and business intelligence. Currently working with Accor S.A., a global hospitality leader, driving data initiatives across enterprise systems. Skilled in Python, Machine Learning, and SQL with a focus on hospitality and e-commerce analytics. Adept at translating complex data findings into clear narratives for both technical and non-technical stakeholders.

EXPERIENCE SUMMARY

- 6+ years of overall experience spanning Technical Support, data analysis, data science and project coordination across global clients including Accor S.A. and Bosch.
- Hands-on expertise in Python, SQL, Machine Learning (scikit-learn, TensorFlow), and data pipeline development for large-scale enterprise environments.
- Proven track record of building and delivering end-to-end data solutions — from data cleaning and feature engineering to model deployment and stakeholder reporting.
- Experience working with structured and unstructured datasets in hospitality, manufacturing, and finance domains.
- Strong analytical and communication skills with experience presenting data insights through dashboards, reports, and executive summaries.
- Agile with experience managing cross-functional teams, SOW documentation, and delivery milestones in multi-vendor environments.
- Demonstrated ability to bridge the gap between business requirements and technical data solutions, reducing reporting cycle time and improving data accuracy.

TECHNICAL SKILLS

Languages: Python, R, SQL

ML / AI: Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn

Data & BI: Power BI, Excel, SEMrush

Tools & Frameworks: Visual Studio, Git, Agile / Scrum

PROJECTS

Accor S.A. — Hospitality

Guest Loyalty Segmentation & Personalisation Engine

Developed a guest segmentation model using clustering (K-Means + RFM analysis) on Accor's ALL loyalty programme data spanning 5,000+ properties globally. Segmented guests into high-value, at-risk, and dormant cohorts, enabling targeted retention campaigns. Integrated segment outputs into CRM dashboards, contributing to a measurable improvement in repeat booking rates for prioritised cohorts.

Tags: Python · K-Means · RFM Analysis · scikit-learn · SQL · Power BI

Dynamic Room Rate Forecasting for Revenue Optimisation

Built a time-series forecasting model to predict hotel occupancy and optimal room pricing across Accor's European and Asia-Pacific properties. Combined historical booking data, seasonal trends, local events, and competitor pricing signals using an LSTM-based pipeline. Model outputs were surfaced to revenue managers via an automated weekly report, reducing manual

forecasting effort by ~40%.

Tags: Python · LSTM · TensorFlow · Time-series · SQL · Power BI

Bosch Global Software — Manufacturing & Supply Chain

Predictive Maintenance Model for Manufacturing Equipment

Designed and built a predictive maintenance classifier to detect early signs of equipment failure on Bosch's manufacturing lines using sensor telemetry data. Applied Random Forest and Gradient Boosting models on multivariate time-series data, achieving 91% precision in failure prediction. Integration with the existing data pipeline reduced unplanned downtime by identifying at-risk equipment 48 hours in advance.

Tags: Python · Random Forest · Time-series · Pandas · SQL

EDI Data Quality & Anomaly Detection Pipeline

Built an automated anomaly detection pipeline to identify data quality issues in Bosch's inbound and outbound EDI interfaces. Used Isolation Forest and rule-based validation to flag malformed records, duplicate transactions, and schema mismatches across 20+ integration endpoints. Reduced manual data quality checks by 60% and improved upstream data reliability for downstream analytics teams.

Tags: Python · Isolation Forest · scikit-learn · SQL · XSD Validation

Hewlett Packard Enterprise — Supply Chain & Logistics

Logistics Delay Analysis & Root Cause Detection using Python

Built a Python-based analysis workflow to identify root causes behind recurring shipment delays and order fulfilment issues across supply chain and logistics operations. Replaced manual Excel-based tracking with automated Python scripts for data cleaning, aggregation, and trend analysis — surfacing patterns by carrier, region, and product category that were previously difficult to spot manually. Findings were shared with planning and logistics teams to support faster issue resolution.

Tags: Python · Pandas · NumPy · Matplotlib · Seaborn · EDA

EXPERIENCE

Associate IT Consultant · ITC Infotech · Client: Accor S.A. Jul 2024 – Present

- Developed a guest segmentation model using K-Means clustering and RFM analysis on Accor's ALL loyalty programme data across 5,000+ properties, identifying high-value, at-risk, and dormant guest cohorts.
- Built an LSTM-based time-series forecasting pipeline to predict hotel occupancy and optimal room pricing, incorporating seasonal trends, local events, and competitor pricing signals.
- Integrated guest segmentation outputs into CRM dashboards using Power BI, enabling targeted retention campaigns and contributing to improved repeat booking rates.
- Automated weekly revenue forecasting reports for revenue managers, reducing manual forecasting effort by approximately 40%.

SEO Analyst (Freelance) · Alice Blue Nov 2023 – Jul 2024

- Drove a 110% increase in organic traffic through data-driven content strategy, keyword analysis, and URL optimisation.
- Leveraged Moz and SEMrush to conduct keyword research, competitor analysis, and performance tracking across 100+ published articles.
- Built end-to-end analytics pipelines to track partner link performance and traffic quality; achieved top-10 Google rankings for 50+ articles.

Data Science · Bosch Global Software (via Klaus IT Solutions) · Bengaluru Mar 2022 – Oct 2023

- Built a predictive maintenance classifier using Random Forest and Gradient Boosting on multivariate sensor telemetry data, achieving 91% precision in equipment failure prediction.
- Engineered time-series features from manufacturing sensor data to flag at-risk equipment up to 48 hours in advance, reducing unplanned downtime.

- Reduced manual data quality checks by 60% by integrating the anomaly detection pipeline into existing data workflows, improving upstream data reliability for downstream analytics teams.
- Performed exploratory data analysis and feature engineering on large-scale manufacturing and supply chain datasets using Python, Pandas, and SQL.

Data Analyst · Hewlett Packard Enterprise (via Bhilwara Technology) · Bengaluru *Apr 2020 – Dec 2021*

- Used Python (Pandas, NumPy) to clean, merge, and standardise shipment and order data pulled from multiple weekly reports.
- Performed exploratory data analysis to identify delay patterns by carrier, region, and product category, replacing manual Excel review.
- Visualized trends using Matplotlib/Seaborn to highlight top contributing factors behind recurring escalations.
- Presented data-backed root cause findings to supply chain and logistics teams, supporting faster, more targeted issue resolution.
- Reduced manual reporting effort by automating recurring trend calculations that were previously done by hand in Excel.

Technical Support Analyst · Hewlett Packard Enterprise (via Bhilwara Technology) · Bengaluru *Mar 2019 – Apr 2020*

- Coordinated across supply chain, logistics, and client service teams to identify root causes of operational issues and drive timely resolution.
- Maintained structured tracking and reporting of issues, contributing to process improvement initiatives.

EDUCATION

Bachelor of Commerce · Jain College of BBA BCA and Commerce, Belgaum *2018*